Cumulative Exam, Organic Chemistry, 100 points
total January 25, 2019

1) How would you use NMR spectroscopy to distinguish between the following pairs of compounds? – be specific in your statements (**20 points total**)

(a)

(b)

2) The following group of signals shows up in your ^{1}H NMR spectrum. Is it possible to assign these signals to an ethyl group? – provide a reason for your statement (**10 points**)

3) A compound with the molecular formula C_4H_8O has a strong IR absorption at 1725 cm^{-1}. Its mass spectrum includes peaks at m/z 44 (base peak) and m/z 29. Propose a structure for the compound and write fragmentation equations showing how peaks having these m/z values arise. **(20 points)**

4) Each of the given compounds$_4$ exhibits a ^{1}H NMR spectrum with only one signal. Deduce their structures. **(30 points)**

(a) C_5H_{10} (b) $C_{12}H_{18}$ (c) $C_5H_8Cl_4$

5) Propose a structure that is consistent with the given set of ^{1}H NMR data. (**20 points**)

C_8H_9Cl	δ(ppm)	splitting	integration
	7.35	m	5H
	5.15	q	1H
	2.00	d	3H

How would the structure change, if the signals a 5.15 and 2.0 ppm were triplets? (5 points)